

NOUPOORT, South Africa

WMO: 686330

Lat: 31.1864S

Lon: 24.9603E

Elev: 4905

StdP: 12.27

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.9	30.4	4.6	8.4	53.1	8.2	10.0	57.1	23.4	52.8	21.7	51.5	2.1	130	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	90.7	57.9	87.9	57.6	85.4	57.3	65.7	75.1	64.6	74.3	63.5	73.4	12.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	103.6	69.0	61.7	98.9	68.3	60.4	94.4	67.6	33.5	74.5	32.5	74.3	31.6	73.5	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	21.0	18.8	DB	22.4	96.6	2.8	6.3	20.4	101.2	18.8	104.8	17.2	108.4	15.2	112.9	(4)
(5)			WB	19.2	68.0	2.1	2.7	17.7	70.0	16.5	71.5	15.3	73.0	13.8	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.0	71.3	70.9	66.7	59.6	53.2	46.7	46.3	50.5	56.6	62.6	65.8	70.2	(6)	
	DBStd	10.78	5.76	4.79	5.72	6.33	5.96	4.84	5.64	6.54	7.20	7.18	7.05	5.33	(7)	
	HDD50	408	0	0	1	8	36	120	137	77	22	5	2	0	(8)	
	HDD65	2687	15	9	49	181	367	550	579	451	265	131	74	16	(9)	
	CDD50	4044	660	586	517	295	135	20	23	91	219	396	476	626	(10)	
	CDD65	850	210	175	101	18	1	0	0	0	12	57	99	177	(11)	
	CDH74	7498	1859	1286	718	134	8	0	0	9	186	596	992	1710	(12)	
	CDH80	2419	689	405	173	19	0	0	0	0	38	165	310	620	(13)	
(14) Wind	WSAvg	8.2	9.5	8.7	8.0	6.9	6.3	6.7	6.8	7.4	8.2	9.6	10.2	10.2	(14)	
(15) Precipitation	PrecAvg	13.1	2.0	2.2	2.1	1.3	0.6	0.6	0.4	0.5	0.4	0.9	1.0	1.5	(15)	
	PrecMax	20.0	4.8	6.6	4.3	3.6	2.4	1.7	1.5	3.0	2.6	4.3	2.9	3.8	(16)	
	PrecMin	8.0	0.1	0.6	0.9	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.0	0.0	(17)	
	PrecStd	3.2	1.3	1.3	0.8	0.8	0.7	0.5	0.4	0.6	0.7	0.9	0.7	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.4	92.1	88.6	81.9	74.9	66.4	67.3	75.1	85.2	89.5	91.0	94.1	(19)	
		MCWB	59.3	58.5	59.7	56.8	52.6	48.5	46.3	49.0	53.0	55.1	56.1	57.7	(20)	
	2%	DB	90.4	88.2	84.6	77.6	71.1	63.1	63.8	70.5	79.3	84.7	86.9	89.7	(21)	
		MCWB	58.8	58.7	58.7	56.7	51.0	46.8	46.0	48.2	50.8	54.1	55.9	57.2	(22)	
	5%	DB	87.0	84.9	81.4	74.3	67.8	60.5	60.9	66.7	75.1	80.7	83.7	86.1	(23)	
		MCWB	58.8	59.2	58.6	55.2	50.3	45.5	45.3	46.9	49.7	53.5	55.4	56.7	(24)	
	10%	DB	83.6	81.7	78.2	70.7	64.2	57.4	57.8	63.0	71.0	76.7	80.1	82.8	(25)	
		MCWB	58.7	59.7	59.0	54.8	49.8	44.3	44.6	45.7	48.7	52.3	54.8	57.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.5	67.5	66.9	63.0	56.7	52.0	50.4	54.6	57.5	60.2	63.9	65.9	(27)	
		MCDB	75.5	77.5	74.5	72.1	64.1	61.1	59.4	64.5	68.6	76.2	71.7	75.0	(28)	
	2%	WB	64.9	66.1	64.8	60.6	54.6	49.5	48.5	52.1	55.2	58.4	61.8	63.6	(29)	
		MCDB	74.3	75.7	74.1	68.8	64.0	57.8	59.0	61.9	70.1	73.8	71.9	74.0	(30)	
	5%	WB	63.8	64.8	63.5	58.9	53.1	47.7	46.8	50.1	53.2	56.8	60.1	62.5	(31)	
		MCDB	73.9	74.7	73.0	67.0	62.5	56.3	57.6	62.4	67.2	71.6	71.7	72.9	(32)	
	10%	WB	62.8	63.6	62.2	57.3	51.5	46.2	45.3	48.0	51.2	55.4	58.5	61.3	(33)	
		MCDB	73.6	73.5	71.9	66.0	61.0	54.6	56.0	60.0	67.3	71.1	71.8	72.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	22.5	21.5	21.6	20.7	21.4	22.3	22.9	23.6	25.7	24.8	24.4	24.5	(35)	
		MCDBR	27.3	26.1	25.9	25.2	25.5	26.0	26.0	27.5	30.8	28.9	28.6	28.7	(36)	
	5% WB	MCWBR	6.8	7.0	8.1	10.5	11.4	13.4	12.9	12.9	13.7	10.3	8.0	7.5	(37)	
		MCWBR	20.0	19.3	19.6	20.2	20.3	21.5	22.8	22.0	24.3	23.5	21.5	21.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.311	0.317	0.306	0.282	0.263	0.250	0.249	0.287	0.313	0.319	0.297	0.300	(40)	
		taud	2.564	2.542	2.551	2.610	2.647	2.651	2.643	2.491	2.396	2.417	2.550	2.586	(41)	
	Ebn at Noon		326	318	312	304	296	294	299	297	304	314	329	331	(42)	
		Edh at Noon	34	34	31	26	23	21	22	29	36	38	34	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2455	2236	1908	1480	1184	1045	1142	1424	1832	2200	2486	2631	(44)	
	RadStd		134	131	91	64	50	58	54	80	96	84	119	134	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page